POKHARA UNIVERSITY

Level: Master
Programme: MSc.CS/ ME.CE
Course: Advanced Computer Architecture

Semester: Spring

Year : 2019
Full Marks: 100
Pass Marks: 60
Time : 4hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions:

1. Differntiate Havard architecture with Von Neumann architecture. Explain the Von Neumann architecture by using a suitable diagram [4+6]
2. What are difference between register indirect addressing mode and indexed addressing mode? Discuss advantages of indexed addressing modes. [6+4]
3. Draw the flowchart of Booths multiplication algorithm and Explain the same with an example. [5+5]
4. Discuss various types of cache mapping with examples. [10]
5. Derive and explain the relation of speedup for normal pipelining. Differentiate super pipelining with super scalar pipelining and normal pipelining with examples. [6+4]
6. You have RAMs 128 X8 and ROM 512 X 8. You need to make RAM of 512 X 8 RAM. Design memory organisation for the given scenario. What can be the total memory accessed in this organization. [10]
7. Discuss various ways of synchronization in distributed system with examples. [4+6]
8. Discuss cloud computing along with its performance challenges, pros and cons. [10]
9. If you were to design 5 address instructions, how would you design it? Justify your arguments with examples. [10]
10. What are different issues related with interconnection of networks and their probable solutions with examples. [10]

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